



Report 4

Breast-cancer-prediction

Dataset = WISCONSIN DIAGNOSIS BREAST CANCER(WDBC)

Friday, 21th july 2024

Content:

Calculating P-Value for each feature with the target values.

Information:

1. Normality Check : Shapiro-Wilk Test
 2. Two-Tailed t-test
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Normality Check :

One factor to use of t-test for a data is, its normality distribution. **Shapiro-Wilk Test** was used to test the normality of the data. The result showed that **none** of the features was normality distributed.

T-test :

The target splitted into two groups of malignant and benign. Then from each of these groups, 200 samples were used to be malignant and benign samples of the target variable. Each of them comprise 10 features. Then the Two-Tailed Test was calculated.

Null-hypothesis :

H0 : There is no significant difference in the means of the feature (e.g., a continuous variable) between the "Malignant" and "Benign" groups.

Alternative Hypothesis (H1 or Ha) :

H1 or Ha : The alternative hypothesis states that there is a significant effect or difference.

Here is the results:

	Features
p_value < 0.05 (There is a significant difference in means)	radius_mean, texture_mean, perimeter_mean, area_mean, smoothness_mean, compactness_mean, concavity_mean, concave points_mean, symmetry_mean, radius_se, perimeter_se, area_se, compactness_se, concavity_se, concave points_se, fractal_dimension_se, radius_worst, texture_worst, perimeter_worst, area_worst, smoothness_worst, compactness_worst, concavity_worst, concave points_worst, symmetry_worst
p_value > 0.05 (There is no significant difference in means)	fractal_dimension_mean, texture_se, smoothness_se, symmetry_se



Results:

1. **25** features were **important** features of the t-test.
2. **5** features were **not important** features of the t-test.